

International Institute of Information Technology, Bangalore
CSE 511 Algorithms: Practice Problems 1
12, August 2026.

1. Let $F(0) = 0, F(1) = 1, F(2) = 2$ and $F(n) = (F(n-1) + F(n-2) + F(n-3) + 1) \% m$. If $n < 10^{10^5}$ and $m < 10^5$, write an efficient algorithm to compute $F(n)$.
2. Let $F(0) = 0, F(1) = 1, F(2) = 2$ and $F(n) = (2F(n-1) - 3F(n-3)) \% m$. If $n < 10^{10^5}$ and $m < 10^5$, write an efficient algorithm to compute $F(n)$.
3. If $T(n) = \Theta(1)$, for $n < 5$, write the solutions to the following recursions, by Masters Theorem.
 - (a) $T(n) = 4T(n/2) + n^2$,
 - (b) $T(n) = 16T(n/2) + n$,
 - (c) $T(n) = 3T(n/3) + n \log n$
 - (d) $T(n) = 2T(n/4) + \log n$
 - (e) $T(n) = 4T(n/2) + n/\log n$
 - (f) $T(n) = 9T(n/3) + n$
 - (g) $T(n) = 3T(n/3) + n^2$
 - (h) $T(n) = 2T(n/4) + n^{2/3}$
 - (i) $T(n) = 3T(n/9) + n^{3/4}$
 - (j) $T(n) = 8T(n/3) + n^2$
 - (k) $T(n) = 3T(n/4) + n \log n$
 - (l) $T(n) = 6T(n/3) + n^2 \log n$
4. What is the complexity of the following algorithms?

(a) $\text{while}(n > 0)\{$
 $\text{for}(i = 1; i < n; i = i * 2) c++;$

 $n = n/2; \}$

(b) $\text{while}(n > 0)\{$
 $\text{for}(i = 1; i < n; i++) c++;$

 $n = n/2; \}$

(c) $j = 1;$
 $\text{while}(j < n)\{$
 $\text{for}(i = 1; i < n; ++i) c ++;$
 $j = 2 * j; \}$

(d) $\text{while}(n > 0)\{$
 $\text{for}(i = 1; i < n; i = i * 3) c ++;$
 $n = n/3; \}$

(e) $\text{while}(n > 0)\{$
 $\text{for}(i = 1; i < n; i ++) c ++;$
 $n = n/3; \}$

(f) $j = 1;$
 $\text{while}(j < n)\{$
 $\text{for}(i = 1; i < n; ++i) c ++;$
 $j = 3 * j; \}$

5. Solution to which of the following recursion is linear ?

- (a) $T(n) = 3T(n/5) + T(n/4) + n$
- (b) $T(n) = 3T(n/9) + 8T(n/11) + n$
- (c) $T(n) = 3T(n/10) + 8T(n/8) + n$
- (d) $T(n) = 3T(n/7) + 4T(n/8) + n$
- (e) $T(n) = 2T(n/5) + 4T(n/7) + n$
- (f) $T(n) = 3T(n/3) + 2T(n/4) + n$
- (g) If $n = 3m, T(n) = n + 5/n \sum_{k=0}^{k=m-1} T(3k)$
- (h) $T(n) = n + 49/n \sum_{k=0}^{k=n/5} T(k)$
- (i) $T(n) = n + 15/n \sum_{k=0}^{k=n/3} T(k)$

6. A binary string is called *dense* if the number of 1's in the string is more than the number of 0's. For example 1, 101, 110101 are *dense*, but 10, 1001, 100001 are not *dense*.

Given a binary string of length n , design an $O(n \log n)$ time algorithm to compute the number of *dense* sub-strings of the given string. For example, given 10101, the answer is 6.

7. Given a binary string of length n , design a linear time algorithm to compute the length of the largest *dense* sub-string of the given string.
8. Given a binary string of length n , design a linear time algorithm to compute the length of the largest sub-string which contains equal number of 0's and 1's.
9. Given a binary string S of length n , design a linear time algorithm to compute k , such that the number of 0's in $S[0..k]$ is equal to number of 1's in $S[k+1..n-1]$.
10. Given an array of sorted integers and an integer $X > 0$, design a linear time algorithm to count the number of pair elements in the array such that $A[j] - A[i] > X$.
11. Given an array of integers, design a $\Theta(n^2)$ algorithm to decide if there is i, j, k such that $A[i] + A[j] = A[k]$.
12. Given an array of integers, design an efficient algorithm to decide if there is i, j, k, l such that $A[i] - 2A[j] = A[k] - 3A[l]$.
13. Given n , radius of a circle with $(0, 0)$ as center, write a linear time algorithm to compute the number of lattice (integer) points inside the circle.
14. Given a stream of n (about 10^9) numbers, design an $O(n)$ time and $O(k)$ space algorithm to find an element of rank k .
15. Given a sequence of n numbers and an integer $k < n$, design a linear time algorithm to find k numbers, closest to the median.
16. Given two sorted arrays of size m and n respectively and an integer k , design an $O(\log k)$ algorithm to find an element of rank k in the merged array.